

Multi agent based incident management system according to ITIL

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Abstract—Various crucial business processes within the enterprise are based on information technology (IT) services and, as consequence, negative IT incidents can interrupt the daily enterprise' activities and cause negative effects such as: a loss of costumers' confidence, loss of productivity and direct financial loss. For that, enterprises look for implementing IT incident management system in order to weaken the negative impact of incidents. However, most of proposed solutions suffer from the complexity of use and integrating, the no-respect of the IT standards and the no-automated incident handling in some phases. In This paper, we propose an ITIL compliant incident management system based on multi-agent technology to overcome the above issues. Also we have enhancing the incidents matchmaking process by a semantic matchmaking algorithm.

Keywords—incident management; ITIL; Multi agent system; Matchmaking algorithm;

I. INTRODUCTION

Nowadays, enterprises have become more and more dependent on IT services to support their business processes and achieve their strategic goals. Managing the enterprise IT more effectively than the competitors is a critical success factor in order to gain costumers confidence and increase the enterprise value. But in practice, managing IT services is not a trivial and straightforward task.

Managing incidents is a crucial area in IT services management; it has as aim to restore normal service operation as quickly as possible following an incident, while minimizing impact to business operations and ensuring quality is maintained. The incident management activities can be divided into two sub phases. The pre-incident phase concerns the prevention and protection techniques in order to anticipate incidents while the post-incident is related to the methods for resolving incidents within established service time. Since we can't ensure the no-incidents with the preventive techniques, the emphasis is on post-incident phase. Several systems have been implemented to this aim, but they don't follow the IT standards guidelines.

To run IT and provide IT services effectively and efficiently, management systems have to follow the best practices standards such as information technology infrastructure library (ITIL). It's a set of processes for IT Service Management, ranging from service level and capacity management to incident management, and to configuration, release and change management [1]. The focus in this paper is

on incident management systems for that we will adopt the bottom-up approach in implementing ITIL processes.

In other side, the technical error in Nasdaq trading system [2] cause thousands of unsatisfied traders and pending law suits, the shutdown of ticketing airport system [3] cause several frustrated travelers. These incidents show clearly how negative IT incidents may cause to lose costumers, reputation and market position. For that, implementing an efficient Incident management system is a primordial task to maintaining the enterprise success.

Most of exiting public incident management systems deal with several issues such as: the complexity of using or maintaining, the interoperability issues in the integration phase, the no-respect to the best practices standards. Our proposition to overcome these issues is the use of multi agent technologies - in order to benefit from their efficiency, robustness and reliability- to implement an incident management system according to ITIL guidelines.

Using ITIL instead of other frameworks is justified by their good reputation in the IT framework market .ITIL is proven and used worldwide since it is flexible and scalable, organizations of all shapes and sizes can implement parts of ITIL to deliver business benefits. Also by adopting such standard, enterprises can gain competitive advantage through value creation and agile change and in the other side increase the customer satisfaction thanks to ITIL reliability.

The remainder of this paper is structured as follows: section two summarizes the state of the art related to the incident management systems in the commercial and non-commercial areas. The third section provides an overview of the proposed incident management system based on multi-agent technology while the last section outlines the conclusion.

II. STATE OF THE ART

An incident is any event which is not part of the standard operation of a service and which may cause an interruption or reduction in the quality of that service. The purpose of Incident Management is to return to the normal service level as soon as possible by mitigating or eliminating the effects of disturbances in IT service [4]. The research effort in this area is divided into two fields the theoretical approaches that offer a set of guidelines to follow during the incident management life cycle; the second field is the free and commercial solutions to the incident management scenario.

The major player in the IT guidelines standards is [5]; it presents a framework of general best practices for efficiently

adopting an IT service management (including IT incident management). According to the latest version of ITIL [6], the incident management lifecycle must includes a set of well defined processes for an effective and efficient handling of incidents; the main steps related on how to record and categorize incidents according to an ITIL checklist incident prioritization is defined in incident logging and categorization process. Additional steps have been added to incident resolution process by 1st level support to explain that incidents should be matched (if possible) to exiting problems and known errors, also a resolution process by 2nd level support have been considerably expanded to provide clearer guidance on when to invoke problem management (i.e. if the underlying cause of an incident cannot be solved with minor change) while the last process is the incident closure and evaluation.

The ITIL guidelines are intentionally no-perspective in order to fit into many kind of organizations. For that, several customized and specified versions of ITIL have been proposed such as [7] which is a customized solution that summarizes the key components of ITIL. Free and lightweight standards family that aims at facilitating the IT service management is proposed in [8]. Another alternative to ITIL is [9], it's a Microsoft solution that provides guidance across the entire IT life cycle and of course the incident management phase is included. Also [10] have proposed a detailed documentation of incident management processes based on best practice.

Unlike the above approaches, some standards focus in specific incident aspect such as the security. Iso[11] and [12] stands out as two of the main standards and guidelines related to information security incident management. Afterwards, structured guidelines based on [11] have been proposed in [13]. It's a quite short and contains just an overview of which activities belongs to each phase. These approaches and standards are proposed to help the IT professionals to manage the incidents effectively and efficiently. But in practice, building an incident management system according to these standards is not an easy and trivial task.

In other side, many commercial software projects for incident management are available. For example, BMC Footprints [14] offers easy-to-use tools for IT service management in mid-sized organizations; Remedy [15] is a configurable and extensible platform for incident tracking while the most prevalent software is NeTSupport [16], it is an online incident and problem management platform according to ITIL. These solutions are web based help desk software and don't cover efficiently the overall incident lifecycle.

To improve incident management process quality, no commercial approaches have been proposed in the academic research field as summarized in table1. These approaches use several technologies and techniques such as data mining and decision making algorithms [17]. Also, Multi agent systems can be a beneficial technology in implementing incident management process thanks to their efficiency, robustness and reliability. However, this technology isn't exploited in the proposed approaches (expect in [18]).

TABLE I. NON COMMERCIAL INCIDENT MANAGEMENT APPROCHES

Approach	MAS	SC?	Scope/limitations
A Multi-agent System for Incident Management Solutions on IT Infrastructures [18].	Yes	NO	A multi agent system for incident management. But it doesn't exploit the MAS benefits in the incident resolution phase and the detection phase is based just on user's report.
SYMIAN: Analysis and performance improvement of the IT incident management process [19].	No	No	SYMIAN, a decision support tool for the performance improvement of the incident management function in IT support organizations/ the adopted strategy can't be applied in high-sized organizations
An Incident Management Model for SaaS Application in the IT Organization [20]	No	Yes (ITIL)	incident management and service restoration tool that support change issues/ lack of information about the services tracking processes
Big Data architecture for IT incident management [21].	No	No	new analysis architecture using Big Data techniques for Incident diagnosis/ time processed to handle an incident request can be the weak point in this architecture
Improving Incident Management Processes in Two IT Service Provider Companies[22].	No	Yes (ITIL)	incident management process to improve customer support activities/ it ignores the pre-incident effects in the improvement process

SC: standards compliance, MAS: multi-agent system

Building an efficient incident management system depends on two factors: following the standards guidelines and using a powerful technology to implement these guidelines. As shown in table1, most of the proposed approaches failed to combine these criteria. Thus, a multi agent system for incident management according to ITIL is proposed in the next section.

III. MULTI-AGENT SYSTEM FOR INCIDENT MANAGEMENT ACCORDING TO ITIL GUIDELINES

A. Motivation of this work

Exiting management systems are divided into two categories. The first one is commercial softwares which are powerful solution for small and mid-sized organizations; however it suffers from the lack of usability and the IT standards compliance. The second one is the no-commercial systems which are proposed by IT researchers. This solution covers the overall incident lifecycle but it suffers from the complexity of using. Also, multi agent technology can plays a crucial role in automating the incident lifecycle but it still doesn't exploited in this area. These issues highlight the need of an approach for automating incident management process using multi agent technology and also in compliance with IT standards guidelines. The proposition and implementation of such approach are the purpose of the next subsection.

B. Proposed approach

One of the promising technologies for the realization of smart systems is multi agent technology. Multi agent system (MAS) is composed of multiple interacting intelligent agents in order to perform tasks and solve problems. In this paper, we propose the use of MAS to handle incidents request, this choice have at aim to benefit from the intrinsic features of MAS - autonomy, decentralization, robustness and flexibility- in the incident management area. Otherwise, approaches that don't respect the best practice standard failed to gain the costumers confidante. For that our approach are based on ITIL guidelines. An overview of this architecture is illustrated in fig1.

According to ITIL, the main activities that our management system supports are:

- *Identification and registration*: concerns the detection and the recording of occurred incidents by the incident management system.
- *Categorization and prioritization*: the incidents handling follows the priority strategy for better utilization of resources and staff support time. For this aim, incidents are categorized by severity, impact and SLA constraints.
- *Diagnosis and escalation*: diagnostic tasks must be performed in order to determine the relevant characteristic of the occurred incident and the adequate incidents support staff.
- *Resolution and recovery*: once the incident solution is found, the recovery tasks are performed.
- *Incident closure*: the end status of the incident is recorded in the closure phase by the system

Since the identification of an incident from the user GUI or the event log by the supervisor agent, an incident request is sent to the incident agent. The role of this agent is recording the incident information and looking for similar previously solved incidents –by invoking matchmaking algorithm- in the incident management data base (IMDB). If the required incident is found, the solution is sent to the requester. In the case where the incident occurs for the first time, the incident information will be sent to the adequate staff by the diagnostic agent according to the severity degree. Once the solution is founded, it's recorded in the (IMDB) and sent to the requester. The main components of this architecture are agents, a detailed agent roles list is illustrated in the next subsections.

User agent is a user system trough the GUI. This interface contains several fields- incident id, incident type, incident source and service affected- to record and monitor incidents. After the fulfilment of this interface with the required information an incident query is generated.

Administrator agent is the responsible of managing the system configuration. It manages the connection and disconnection process, the system resources and the configuration changes.

The main role the *supervisor agent* is to supervise the organization services which are related to the IT in order to detect the abnormal incidents. It's done by visiting the event log database that contains information system errors' activity. The main functionalities of this agent are:

- Supervise the IT services in order to automatically detect incidents.
- If an incident is occurred, the supervisor agent classifies and categorizes this incident according to a specific scheme (application incident, hardware incident, infected services...)
- Send an incident request with the required information to the incident agent.

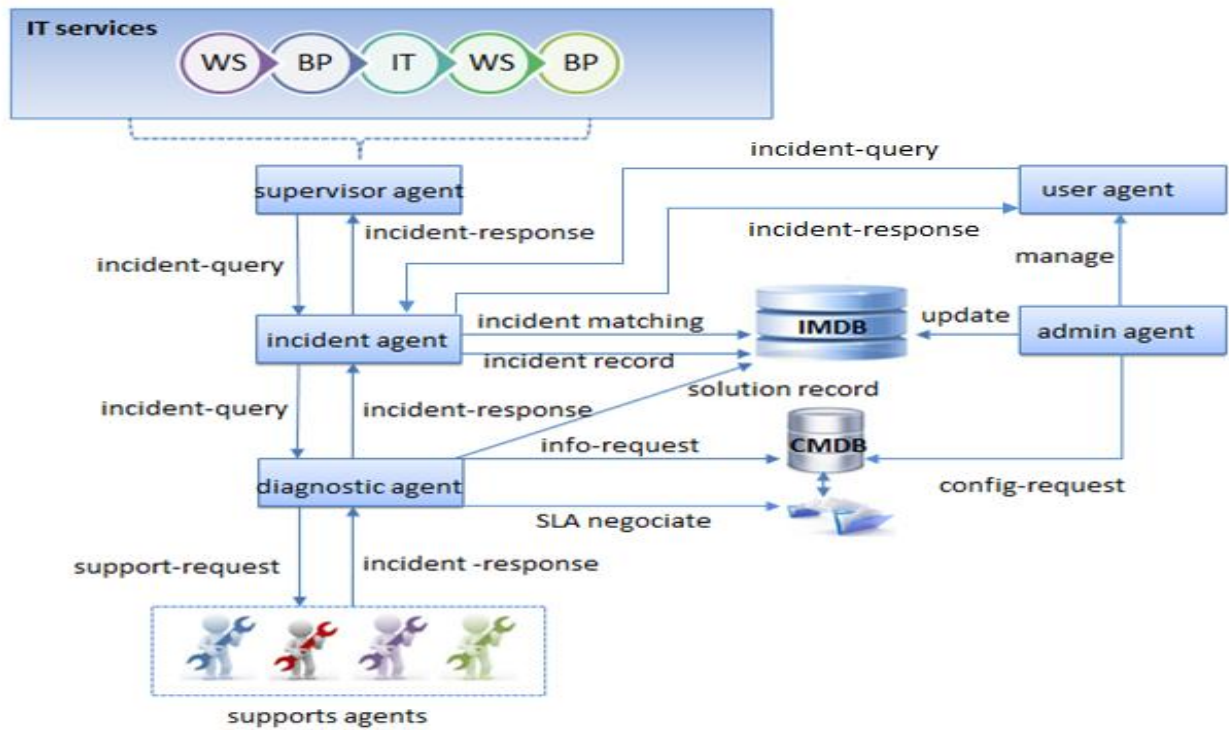
Incident agent aims at handling incident requests which are received from the supervisor agent or the user agent. The main roles of this agent are:

- Record the information related to the incident in the IMDB.
- Look for similar incident in the IMBD. To achieve this goal, we propose a semantically enhanced algorithm to determine the incident similarity' degree.
- If the presented incident is previously occurred and resolved, the incident agent sends the founded solution in IMBD to the requester and calculates the productivity rate. In case of this rate is superior to the fixed threshold by service level agreement (SLA), the problem manager is informed in order to identify the problem source.
- If the presented incident is occurred for the first time, the incident agent sends the collected information to the diagnostic agent.

Diagnostic agent calculates the incident severity degree in function of the impact and emergency degree which is presented in configuration management database (CMDB); also it determines the allowed time average to resolve the incident according to the SLA. Based to these factors, several tasks have been performed by this agent:

- Assign the incident to the adequate support agent according to the incident category and severity.
- Monitor the lifecycle of incidents solving process. Through this agent, we can supervise the performance and the quality of the incident management process by supervising the different states of each incident in the system.

Support agents are invoked when the incident agent failed to solve the incident based on previously resolved incidents. They are grouped in different categories (hardware support, software support, network support...). To each agent, a technical staff is associated according to the agent category.



BP: business process
WS: web service

Fig. 1. Multi agent system for incident management

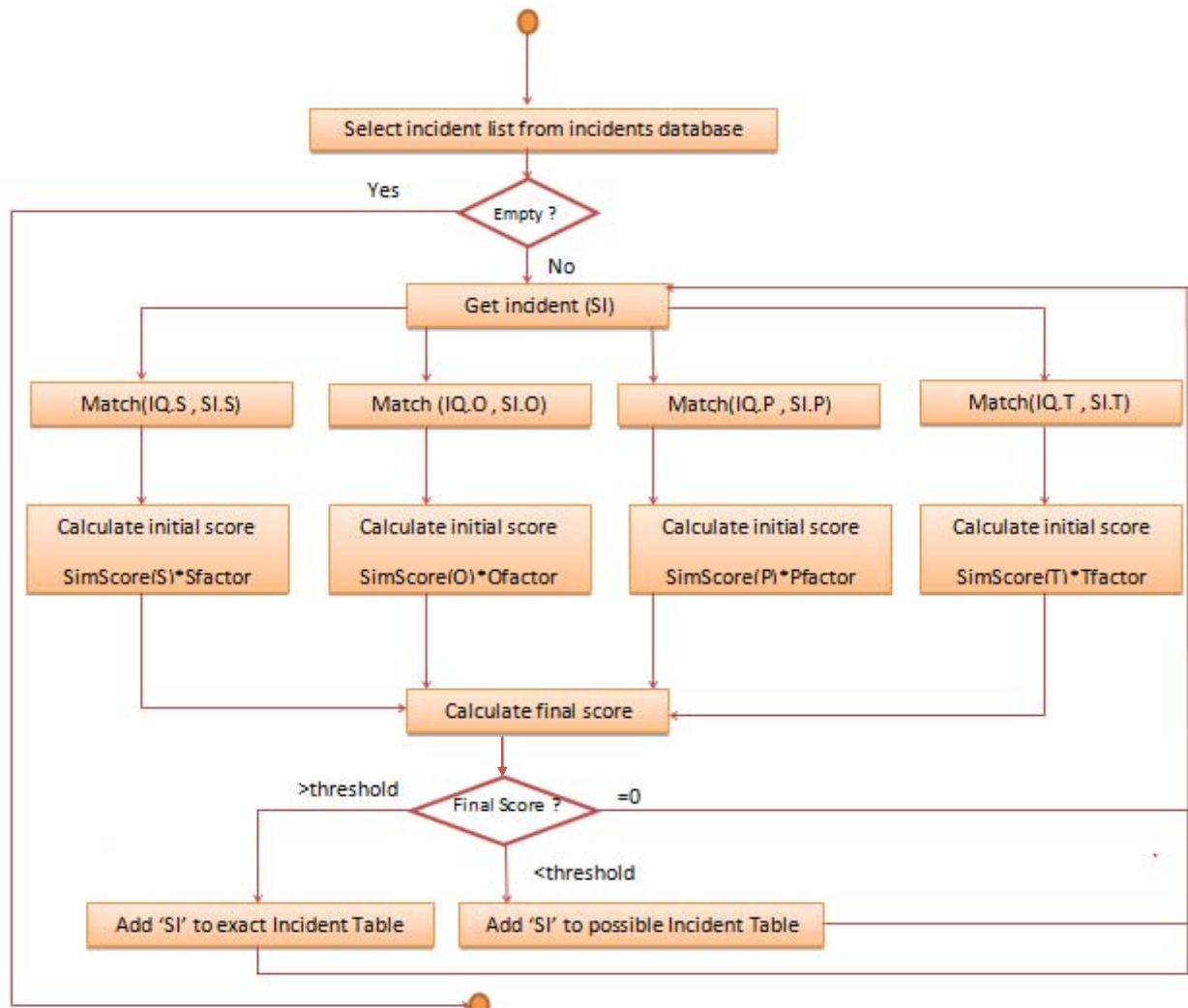


Fig. 2. Semantic incident matchmaking

C. Incident matchmaking process

Once an incident is recorded, the incident agent searches in IMDB for similar incidents previously solved in order to minimize the response-time and the resources consuming. The incidents information included in the IMDB are recorded according to a specific XML representation (ID, Object, Solution, Description...); as shown in fig 2, In the proposed incident matching algorithm-wish is based on our previous work [23] - we are interesting just by the information included in the following tags:

- Object (O): identifies the source of the incident (web browser, printer, excel...).
- Type (T): the type of the object (hardware or application).
- Service(S): determines the affected service by the incident (printing, connexion, mailing...).
- Problem (P): the problem generated by this incident (shut-down, error, fault...).

The matching between incidents is equivalent to the matching between their 4-tuple representations. To perform this task, the matching algorithm is based on semantic matching techniques: i.e. the matching between concepts is not syntactic, but it is based on the relation between these concepts in their OWL (Web Ontology Language). This ontology represents the shared vocabulary for incidents based on previously recorded information of incidents. In other side, the main benefit of this algorithm is the improvement of the accuracy of matching process since it's based on semantic techniques.

The matching algorithm between incidents representation recognizes four degrees: exact matching when two concepts are the same, plugin matching is when concept A includes a concept B while subsume matching is similar with the plugin but the concept A is more generic than the concept B, and the fail matching is when concepts are different.

In case of an incident matching request, the manager agent calculates the similarity degree between the incident-query (IQ) and the selected incident (SI) from the incidents database based on this function:

$$\text{Match}(IQ_i, SI_i) = \begin{cases} 3 & \text{if } IQ_i = SI_i & \text{exact matching} \\ 2 & \text{if } IQ_i \subset SI_i & \text{plugin matching} \\ 1 & \text{if } IQ_i \supset SI_i & \text{subsume matching} \\ 0 & \text{else} & \text{fail matching} \end{cases}$$

Then this matching degree is multiplied by the weight of incident factor (service-Sfactor, Object-Ofactor...), it's fixed to prioritize an incident characteristic instead of other. The final score that represents the similarity degree between these incidents is the sum of the four-tuplet (object, type, service, problem) matching degrees:

$$\text{Similarity}(IQ, SI) = \sum_{i=1}^4 \text{Match}(IQ_i, SI_i)$$

If the final similarity degree is superior to the fixed threshold - it's calculated based on the incident' factors in order to determine the exact matching degree- by the manager agent, the incident query and the selected incident are similar. When the final degree is between 1 and the threshold, the selected incident is a possible candidate to satisfy the incident request (i.e. subsume and plugin matching). In other case, there is no matching between the incident query and the selected incident.

D. Implementaion details

The system has been implemented using JADE (Java Agent Development Framework) from CSELT, Turin, Italy. JADE is a middle-ware that could be used to develop agent-based applications in compliance with the FIPA (Foundation for Intelligent Physical Agents) specifications for inter-operable intelligent multi-agent systems). The incident management information (IMDB) and the configuration information (CMDB) are stored in oracle database. For testing reasons, a simulation environment of governmental agency IT services that contains over 20 IT services is used.

E. Result and discussion

To demonstrate the viability of our system, a comparison with up-to-date similar incident management is required while performing such comparison can't lead to a statistical indicators because the continuously changed incident' environment. Thus, a functionality comparison according to four views is performed as shown in table 2.

TABLE II. TABLE COMPARISON OF INCIDENT MANAGEMENT SYSTEMS

Approach views	A MAS for Incident Management Solutions on IT Infrastructures [18]	Our proposed approach	Footprint IT service management software [14]
Standards compliance	ITIL guidelines are supported	It's an ITIL compliant approach	It's not based on a standards guidelines
Scope	The incident detection phase can be the weak point in this approach since it's not automated and based just on user's queries.	It covers the overall incident lifecycle in an automated way and it can be adapted in a distributed environment	It's specific to a mid-sized organization
usability	A no convivial GUI	Easy to use and maintain	Easy to install and use
Reliability and efficiency	It's a tracking incidents system that doesn't exploit the previously resolved incidents to minimize the time of new incidents resolution	A semantic incident matchmaking phase is added in order to minimize the incident resolution time and increase the technical staff intervention	It's a semi automated solution that support just the tracking of desk services and doesn't support the adding of other IT services

Otherwise, using multi agent technology and semantic matchmaking incident algorithm in our system leads to overcome several lacks and providing several benefits as follows:

- **Autonomy and intelligence:** Incidents management process requires the interaction between several entities in order to perform incidents request. This feature makes the multi agent technology as natural solution for incidents management, Agents are such autonomously and intelligent entities that can communicate in order to handle incidents.
- **Automated processing:** the use of multi-agent systems contributes to the automating of the incidents management lifecycle even in the detection phase by the supervisor agent.
- **Effective Matchmaking process:** the use of semantically enhanced algorithm for incidents matchmaking improves the quality of the new incidents resolving based on previously solved incidents.
- **Abstract design:** The introduction of agents adds new abstract design layers, which results in a more hierarchical and structural design that is easier to comprehend and also to map according to ITIL guidelines.
- **Simplicity and decomposition:** Complex systems are required to manage incidents within high-sized organization. However, MAS provide a simplified solution for system decomposition allowing that the inherent properties of a complex system can be reproduced in a simplified and flexible agent-based system.
- **ITIL compliance:** systems that are implementing without respecting best practice standards fail to gain the costumers and the company confidante.
- **Efficiency:** Time of response to an incident-request using multi-agent system is optimized since they operate in asynchronous and parallel mode with the less consumption of network resources.
- **Reliability and flexibility:** the failure of an agent or the adding of new agents does not affect the operationally of the overall system.

IV. CONCLUSION

Enterprises looks for implementing IT incident management systems in order to increase the recovery speed after an incident is occurred and decrease the negative impact on business quality, how enterprises manage IT services and especially in the case of incidents is one of the key factors of their success. However, Most of existing applications for incident management deal with some issues such as the

complexity of use and integrating, the no-respect of the IT standards guidelines and the no-automated handling.

The aim of our research work is to implement an incident management system that follows the standards guidelines and use powerful technology. We have proposed the use of multi-agent technology in order to benefit from their intrinsic characteristic for automating the incident handling process not just in the incident solving phase but also in the detection phase. In other side, using incidents previously solved information to find solution for new occurred incidents can optimize the use of resources and time. For this aim, we have proposed a semantically enhanced algorithm for improving the incidents matchmaking algorithm.

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